

ZOH

find $G(z)$ we need only observe that the $y(kT)$ are samples of the plant output when the input is from the D/A converter. As for the D/A converter, we assume that this device, commonly called a zero-order hold or ZOH, accepts a sample $u(kT)$ at $t = kT$ and holds its output constant at this value until the next sample is sent at $t = kT + T$. The piecewise constant output of the D/A is the signal, $u(t)$, that is applied to the plant.

Our problem is now really quite simple because we have just seen that the discrete transfer function is the z -transform of the samples of the output when the input samples are the unit pulse at $k = 0$. If $u(kT) = 1$ for $k = 0$ and $u(kT) = 0$ for $k \neq 0$, the output of the D/A converter is a pulse of width T seconds and height 1, as sketched in Fig. 4.13. Mathematically, this pulse is given by $1(t) - 1(t - T)$. Let us call the particular output in response to the pulse shown in Fig. 4.13 $y_1(t)$. This response is the difference between the step response [to $1(t)$] and the delayed step response [to $1(t - T)$]. The Laplace transform of the step response is $G(s)/s$. Thus in the transform domain the unit pulse response of the plant is

$$Y_1(s) = (1 - e^{-Ts}) \frac{G(s)}{s}, \quad (4.40)$$

and the required transfer function is the z -transform of the samples of the inverse of $Y_1(s)$, which can be expressed as

$$\begin{aligned} G(z) &= \mathcal{Z}\{Y_1(kT)\} \\ &= \mathcal{Z}\{\mathcal{L}^{-1}\{Y_1(s)\}\} \triangleq \mathcal{Z}\{Y_1(s)\} \\ &= \mathcal{Z}\{(1 - e^{-Ts}) \frac{G(s)}{s}\}. \end{aligned}$$

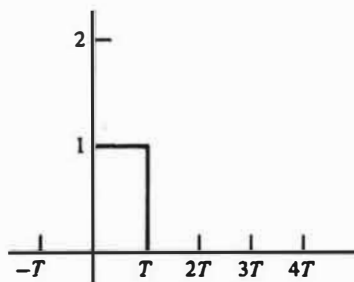
This is the sum of two parts. The first part is $\mathcal{Z}\{\frac{G(s)}{s}\}$, and the second is

$$\mathcal{Z}\{e^{-Ts} \frac{G(s)}{s}\} = z^{-1} \mathcal{Z}\{\frac{G(s)}{s}\}$$

because e^{-Ts} is exactly a delay of one period. Thus the transfer function is

$$G(z) = (1 - z^{-1}) \mathcal{Z}\left\{\frac{G(s)}{s}\right\}. \quad (4.41)$$

Figure 4.13
D/A output for
unit-pulse input



◆ **Example 4.5** *Discrete Transfer Function of 1st-Order System*

What is the discrete transfer function of

$$G(s) = a/(s + a)$$

preceded by a ZOH?

Solution. We will apply the formula (4.41)

$$\frac{G(s)}{s} = \frac{a}{s(s+a)} = \frac{1}{s} - \frac{1}{s+a},$$

and the corresponding time function is

$$\mathcal{L}^{-1} \left\{ \frac{G(s)}{s} \right\} = 1(t) - e^{-at} 1(t).$$

The samples of this signal are $1(kT) - e^{-akT} 1(kT)$, and the z -transform of these samples is

$$\begin{aligned} \mathcal{Z} \left\{ \frac{G(s)}{s} \right\} &= \frac{z}{z-1} - \frac{z}{z-e^{-aT}} \\ &= \frac{z(1-e^{-aT})}{(z-1)(z-e^{-aT})}. \end{aligned}$$

We could have gone to the tables in Appendix B and found this result directly as Entry 12. Now we can compute the desired transform as

$$\begin{aligned} G(z) &= \frac{z-1}{z} \frac{z(1-e^{-aT})}{(z-1)(z-e^{-aT})} \\ &= \frac{1-e^{-aT}}{z-e^{-aT}}. \end{aligned} \tag{4.42}$$

◆ **Example 4.6** *Discrete Transfer Function of a $1/s^2$ Plant*

What is the discrete transfer function of

$$G(s) = \frac{1}{s^2}$$

preceded by a ZOH?

Solution. We have

$$G(z) = (1 - z^{-1}) \mathcal{Z} \left\{ \frac{1}{s^3} \right\}.$$

This time we refer to the tables in Appendix B and find that the z -transform associated with $1/s^3$ is

$$\frac{T^2}{2} \frac{z(z+1)}{(z-1)^3},$$

and therefore Eq. (4.41) shows that

$$G(z) = \frac{T^2(z+1)}{2(z-1)^2}. \quad (4.43)$$

The MATLAB function, `c2d.m` computes Eq. (4.41) (the ZOH method is the default) as well as other discrete equivalents discussed in Chapter 6. It is able to accept the system in any of the forms.

◆ **Example 4.7** *Discrete Transfer Function of a $1/s^2$ Plant Using MATLAB*

Use MATLAB to find the discrete transfer function of

$$G(s) = \frac{1}{s^2}$$

preceded by a ZOH, assuming the sample period is $T = 1$ sec.

Solution. The MATLAB script

```
T = 1;
s=tf('s');
sysC = 1/s^2;
sysD = c2d(sysC,T,'zoh')
```

produces the result that

$$G(z) = \frac{0z^2 + 0.5z + 0.5}{z^2 - 2z + 1} = 0.5 \frac{z+1}{(z-1)^2}$$

which is the same as Eq. (4.43) with $T = 1$.

4.3.2 *Continuous Time Delay

We now consider computing the discrete transfer function of a continuous system preceded by a ZOH with pure time delay. The responses of many chemical process-control plants exhibit pure time delay because there is a finite time of transport of fluids or materials between the process and the controls and/or the

sensors. Also, we must often consider finite computation time in the digital controller, and this is exactly the same as if the process had a pure time delay. With the techniques we have developed here, it is possible to obtain the discrete transfer function of such processes exactly, as Example 4.8 illustrates.

◆ **Example 4.8** *Discrete Transfer Function of 1st-Order System with Delay*

Find the discrete transfer function of the mixer in Appendix A.3 with $a = 1$, $T = 1$, and $\lambda = 1.5$.

Solution. The fluid mixer problem in Appendix A.3 is described by

$$G(s) = e^{-\lambda s} H(s).$$

The term $e^{-\lambda s}$ represents the delay of λ seconds, which includes both the process delay and the computation delay, if any. We assume that $H(s)$ is a rational transfer function. To prepare this function for computation of the z -transform, we first define an integer ℓ and a positive number m less than 1.0 such that $\lambda = \ell T - mT$. With these definitions we can write

$$\frac{G(s)}{s} = e^{-\ell Ts} \frac{e^{mTs} H(s)}{s}.$$

Because ℓ is an integer, this term reduces to $z^{-\ell}$ when we take the z -transform. Because $m < 1$, the transform of the other term is quite direct. We select $H(s) = a/(s + a)$ and, after the partial fraction expansion of $H(s)/s$, we have

$$G(z) = \frac{z-1}{z^{\ell+1}} \mathcal{Z} \left\{ \frac{e^{mTs}}{s} - \frac{e^{mTs}}{s+a} \right\}.$$

To complete the transfer function, we need the z -transforms of the inverses of the terms in the braces. The first term is a unit step shifted left by mT seconds, and the second term is an exponential shifted left by the same amount. Because $m < 1$, these shifts are less than one full period, and no sample is picked up in negative time. The signals are sketched in Fig. 4.14.

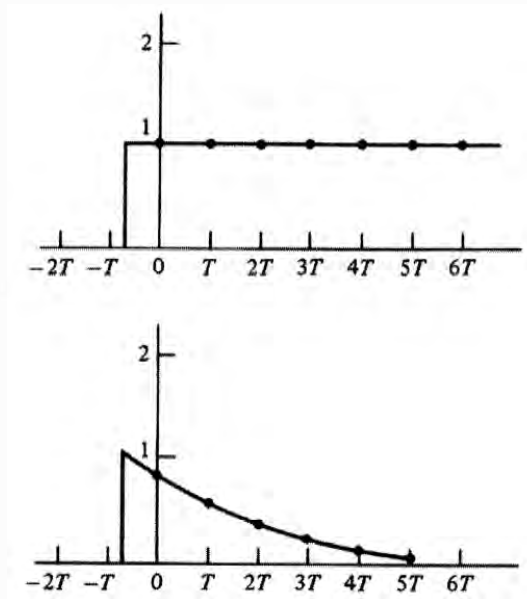
The samples are given by $1(kT)$ and $e^{-aT(k+m)}1(kT)$. The corresponding z -transforms are $z/(z-1)$ and $ze^{-amT}/(z-e^{-aT})$. Consequently the final transfer function is

$$\begin{aligned} G(z) &= \frac{z-1}{z} \frac{1}{z^{\ell}} \left\{ \frac{z}{z-1} - \frac{ze^{-amT}}{z-e^{-aT}} \right\} \\ &= \frac{z-1}{z^{\ell}} \left\{ \frac{z-e^{-aT}-(z-1)e^{-amT}}{(z-1)(z-e^{-aT})} \right\} \\ &= (1-e^{-amT}) \frac{z+\alpha}{z^{\ell}(z-e^{-aT})} \end{aligned}$$

where the zero position is at $-\alpha = -(e^{-amT} - e^{-aT})/(1 - e^{-amT})$. Notice that this zero is near the origin of the z -plane when m is near 1 and moves outside the unit circle to near $-\infty$ when m approaches 0. For specific values of the mixer, we take $a = 1$, $T = 1$, and $\lambda = 1.5$. Then we can compute that $\ell = 2$ and $m = 0.5$. For these values, we get

$$G(z) = \frac{0.3935(z + 0.6025)}{z^2(z - 0.3679)} \quad (4.44)$$

Figure 4.14
Sketch of the shifted
signals showing sample
points



In MATLAB, the transfer function for this system would be computed by

```
Td = 1.5,  a = 1,  T = 1
sysC = tf(a,[1  a], 'OutputDelay',Td)
sysD = c2d(sysC,T,zoh)
```

4.3.3 State-Space Form

Computing the z-transform using the Laplace transform as in Eq. (4.41) is a very tedious business that is unnecessary with the availability of computers. We will next develop a formula using state descriptions that moves the tedium to the computer. A continuous, linear, constant-coefficient system of differential equations was expressed in Eq. (2.1) as a set of first-order matrix differential equations. For a scalar input, it becomes

$$\dot{\mathbf{x}} = \mathbf{F}\mathbf{x} + \mathbf{G}u + \mathbf{G}_1w, \quad (4.45)$$

where u is the scalar control input to the system and w is a scalar disturbance input. The output was expressed in Eq. (2.2) as a linear combination of the state, $\mathbf{x}$, and the input, u , which becomes for scalar output

$$y = \mathbf{H}\mathbf{x} + Ju. \quad (4.46)$$

Often the sampled-data system being described is the plant of a control problem, and the parameter J in Eq. (4.46) is zero and will frequently be omitted.

◆ Example 4.9 State Representation of a $1/s^2$ Plant

Apply Eqs. (4.45) and (4.46) to the double integrator plant of the satellite control problem in Appendix A.1

$$G(s) = \frac{1}{s^2}.$$

Solution. The satellite attitude-control example is shown in block diagram form in Fig. 4.15 and the attitude (θ) and attitude rate ($\dot{\theta}$) are defined to be x_1 and x_2 , respectively. Therefore, the equations of motion can be written as

$$\begin{aligned} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_{\mathbf{F}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\mathbf{G}} u, \\ \theta = y &= \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{\mathbf{H}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned} \quad (4.47)$$

which, in this case, turns out to be a rather involved way of writing

$$\ddot{\theta} = u.$$

The representations given by Eqs. (4.45) and (4.46) are not unique. Given one state representation, any nonsingular linear transformation of that state such as $Bx = Tx$ is also an allowable alternative realization of the same system.

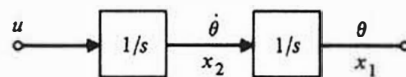
If we let $\xi = Tx$ in Eqs. (4.45) and (4.46), we find

$$\begin{aligned} \dot{\xi} &= T\dot{x} = T(\mathbf{F}x + \mathbf{G}u + \mathbf{G}_1 w) \\ &= T\mathbf{F}x + T\mathbf{G}u + T\mathbf{G}_1 w, \\ \dot{\xi} &= T\mathbf{F}T^{-1}\xi + T\mathbf{G}u + T\mathbf{G}_1 w, \\ y &= \mathbf{H}T^{-1}\xi + Ju. \end{aligned}$$

If we designate the system matrices for the new state ξ as $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$, then

$$\dot{\xi} = \mathbf{A}\xi + \mathbf{B}u + \mathbf{B}_1 w, \quad y = \mathbf{C}\xi + \mathbf{D}u,$$

Figure 4.15
Satellite attitude control
in classical
representation



where

$$\mathbf{A} = \mathbf{T}\mathbf{F}\mathbf{T}^{-1}, \quad \mathbf{B} = \mathbf{T}\mathbf{G}, \quad \mathbf{B}_1 = \mathbf{T}\mathbf{G}_1, \quad \mathbf{C} = \mathbf{H}\mathbf{T}^{-1}, \quad D = J.$$

◆ **Example 4.10** *State Transformation for $1/s^2$ Plant*

Find the state representation for the case with the state definitions of the previous example interchanged.

Solution. Let $\xi_1 = x_2$ and $\xi_2 = x_1$ in Eq. (4.47); or, in matrix notation, the transformation to interchange the states is

$$\mathbf{T} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

In this case $\mathbf{T}^{-1} = \mathbf{T}$, and application of the transformation equations to the system matrices of Eq. (4.47) gives

$$\mathbf{A} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 0 & 1 \end{bmatrix}.$$

Most often, a change of state is made to bring the description matrices into a useful canonical form. We saw earlier how a single high-order difference equation could be represented by a state description in control or in observer canonical form. Also, there is a very useful state description corresponding to the partial-fraction expansion of a transfer function. State transformations can take a general description for either a continuous or a discrete system and, subject to some technical restrictions, convert it into a description in one or the other of these forms, as needed.

We wish to use the state description to establish a general method for obtaining the difference equations that represent the behavior of the continuous plant. Fig. 4.16 again depicts the portion of our system under consideration. Ultimately, the digital controller will take the samples $y(k)$, operate on that sequence by means of a difference equation, and put out a sequence of numbers, $u(k)$, which are the inputs to the plant. The loop will, therefore, be closed. To analyze the result, we must be able to relate the samples of the output $y(k)$ to the samples of the control $u(k)$. To do this, we must solve Eq. (4.45).

We will solve the general equation in two steps. We begin by solving the equation with only initial conditions and no external input. This is the homogeneous equation

$$\dot{\mathbf{x}}_h = \mathbf{F}\mathbf{x}_h(t), \quad \mathbf{x}_h(t_0) = \mathbf{x}_0. \quad (4.48)$$

Figure 4.16
System definition with
sampling operations
shown



To solve this, we assume the solution is sufficiently smooth that a series expansion of the solution is possible

$$\mathbf{x}_h(t) = \mathbf{A}_0 + \mathbf{A}_1(t - t_0) + \mathbf{A}_2(t - t_0)^2 + \dots \quad (4.49)$$

If we let $t = t_0$, we find immediately that $\mathbf{A}_0 = \mathbf{x}_0$. If we differentiate Eq. (4.49) and substitute into Eq. (4.48), we have

$$\mathbf{A}_1 + 2\mathbf{A}_2(t - t_0) + 3\mathbf{A}_3(t - t_0)^2 + \dots = \mathbf{F}\mathbf{x}_h$$

and, at $t = t_0$, $\mathbf{A}_1 = \mathbf{F}\mathbf{x}_0$. Now we continue to differentiate the series and the differential equation and equate them at t_0 to arrive at the series

$$\mathbf{x}_h(t) = \left[\mathbf{I} + \mathbf{F}(t - t_0) + \frac{\mathbf{F}^2(t - t_0)^2}{2} + \frac{\mathbf{F}^3(t - t_0)^3}{6} + \dots \right] \mathbf{x}_0.$$

This series is defined as the matrix exponential and written

$$\mathbf{x}_h(t) = e^{\mathbf{F}(t-t_0)} \mathbf{x}(t_0), \quad (4.50)$$

where, by definition, the matrix exponential is

$$\begin{aligned} e^{\mathbf{F}(t-t_0)} &= \mathbf{I} + \mathbf{F}(t - t_0) + \mathbf{F}^2 \frac{(t - t_0)^2}{2!} + \mathbf{F}^3 \frac{(t - t_0)^3}{3!} + \dots \\ &= \sum_{k=0}^{\infty} \mathbf{F}^k \frac{(t - t_0)^k}{k!}. \end{aligned} \quad (4.51)$$

It can be shown that the solution given by Eq. (4.50) is unique, which leads to very interesting properties of the matrix exponential. For example, consider two values of t : t_1 and t_2 . We have

$$\mathbf{x}(t_1) = e^{\mathbf{F}(t_1-t_0)} \mathbf{x}(t_0)$$

and

$$\mathbf{x}(t_2) = e^{\mathbf{F}(t_2-t_0)} \mathbf{x}(t_0).$$

Because t_0 is arbitrary also, we can express $\mathbf{x}(t_2)$ as if the equation solution began at t_1 , for which

$$\mathbf{x}(t_2) = e^{\mathbf{F}(t_2-t_1)} \mathbf{x}(t_1).$$

Substituting for $\mathbf{x}(t_1)$ gives

$$\mathbf{x}(t_2) = e^{\mathbf{F}(t_2-t_1)} e^{\mathbf{F}(t_1-t_0)} \mathbf{x}(t_0).$$

We now have two separate expressions for $\mathbf{x}(t_2)$, and, if the solution is unique, these must be the same. Hence we conclude that

$$e^{\mathbf{F}(t_2-t_0)} = e^{\mathbf{F}(t_2-t_1)} e^{\mathbf{F}(t_1-t_0)} \quad (4.52)$$

for all t_2, t_1, t_0 . Note especially that if $t_2 = t_0$, then

$$\mathbf{I} = e^{-\mathbf{F}(t_1-t_0)} e^{\mathbf{F}(t_1-t_0)}.$$

Thus we can obtain the inverse of $e^{\mathbf{F}t}$ by merely changing the sign of t ! We will use this result in computing the particular solution to Eq. (4.45).

The particular solution when u is not zero is obtained by using the method of **variation of parameters**.¹¹ We guess the solution to be in the form

$$\mathbf{x}_p(t) = e^{\mathbf{F}(t-t_0)} \mathbf{v}(t), \quad (4.53)$$

where $\mathbf{v}(t)$ is a vector of variable parameters to be determined [as contrasted to the constant parameters $\mathbf{x}(t_0)$ in Eq. (4.50)]. Substituting Eq. (4.53) into Eq. (4.45), we obtain

$$\mathbf{F} e^{\mathbf{F}(t-t_0)} \mathbf{v} + e^{\mathbf{F}(t-t_0)} \dot{\mathbf{v}} = \mathbf{F} e^{\mathbf{F}(t-t_0)} \mathbf{v} + \mathbf{G} u.$$

and, using the fact that the inverse is found by changing the sign of the exponent, we can solve for $\dot{\mathbf{v}}$ as

$$\dot{\mathbf{v}}(t) = e^{-\mathbf{F}(t-t_0)} \mathbf{G} u(t).$$

Assuming that the control $u(t)$ is zero for $t < t_0$, we can integrate $\dot{\mathbf{v}}$ from t_0 to t to obtain

$$\mathbf{v}(t) = \int_{t_0}^t e^{-\mathbf{F}(t-\tau)} \mathbf{G} u(\tau) d\tau.$$

Hence, from Eq. (4.53), we get

$$\mathbf{x}_p(t) = e^{\mathbf{F}(t-t_0)} \int_{t_0}^t e^{-\mathbf{F}(t-\tau)} \mathbf{G} u(\tau) d\tau,$$

and simplifying, using the results of Eq. (4.52), we obtain the particular solution (convolution)

$$\mathbf{x}_p(t) = \int_{t_0}^t e^{\mathbf{F}(t-\tau)} \mathbf{G} u(\tau) d\tau. \quad (4.54)$$

The total solution for $w = 0$ and $u \neq 0$ is the sum of Eqs. (4.50) and (4.54):

$$\mathbf{x}(t) = e^{\mathbf{F}(t-t_0)} \mathbf{x}(t_0) + \int_{t_0}^t e^{\mathbf{F}(t-\tau)} \mathbf{G} u(\tau) d\tau. \quad (4.55)$$

¹¹ Due to Joseph Louis Lagrange, French mathematician (1736–1813). We assume $w = 0$, but because the equations are linear, the effect of w can be added later.

We wish to use this solution over one sample period to obtain a difference equation: hence we juggle the notation a bit (let $t = kT + T$ and $t_0 = kT$) and arrive at a particular version of Eq. (4.55):

$$\mathbf{x}(kT + T) = e^{\mathbf{F}T} \mathbf{x}(kT) + \int_{kT}^{kT+T} e^{\mathbf{F}(kT+T-\tau)} \mathbf{G}u(\tau) d\tau. \quad (4.56)$$

This result is not dependent on the type of hold because u is specified in terms of its continuous time history, $u(t)$, over the sample interval. A common and typically valid assumption is that of a zero-order hold (ZOH) with no delay, that is,

$$u(\tau) = u(kT), \quad kT \leq \tau < kT + T.$$

If some other hold is implemented or if there is a delay between the application of the control from the ZOH and the sample point, this fact can be accounted for in the evaluation of the integral in Eq. (4.56). The equations for a delayed ZOH will be given in the next subsection. To facilitate the solution of Eq. (4.56) for a ZOH with no delay, we change variables in the integral from τ to η such that

$$\eta = kT + T - \tau.$$

Then we have

$$\mathbf{x}(kT + T) = e^{\mathbf{F}T} \mathbf{x}(kT) + \int_0^T e^{\mathbf{F}\eta} d\eta \mathbf{G}u(kT). \quad (4.57)$$

If we define

$$\begin{aligned} \Phi &= e^{\mathbf{F}T} \\ \Gamma &= \int_0^T e^{\mathbf{F}\eta} d\eta \mathbf{G}, \end{aligned} \quad (4.58)$$

Eqs. (4.57) and (4.46) reduce to difference equations in standard form

$$\begin{aligned} \mathbf{x}(k+1) &= \Phi \mathbf{x}(k) + \Gamma u(k) + \Gamma_1 w(k), \\ y(k) &= \mathbf{H} \mathbf{x}(k), \end{aligned} \quad (4.59)$$

where we include the effect of an impulsive or piecewise constant disturbance, w , and assume that $J = 0$ in this case. If w is a constant, then Γ_1 is given by Eq. (4.58) with $\mathbf{G}$ replaced by $\mathbf{G}_1$. If w is an impulse, then $\Gamma_1 = \mathbf{G}_1$.¹² The Φ series expansion

$$\Phi = e^{\mathbf{F}T} = \mathbf{I} + \mathbf{F}T + \frac{\mathbf{F}^2 T^2}{2!} + \frac{\mathbf{F}^3 T^3}{3!} + \dots$$

¹² If $w(t)$ varies significantly between its sample values, then an integral like that of Eq. (4.56) is required to describe its influence on $\mathbf{x}(k+1)$. Random disturbances are treated in Chapter 9.

can also be written

$$\Phi = I + FT\Psi, \quad (4.60)$$

where

$$\Psi = I + \frac{FT}{2!} + \frac{F^2 T^2}{3!} + \dots$$

The Γ integral in Eq. (4.58) can be evaluated term by term to give

$$\begin{aligned} \Gamma &= \sum_{k=0}^{\infty} \frac{F^k T^{k+1}}{(k+1)!} G \\ &= \sum_{k=0}^{\infty} \frac{F^k T^k}{(k+1)!} TG \\ &= \Psi TG. \end{aligned} \quad (4.61)$$

We evaluate Ψ by a series in the form

$$\Psi \approx I + \frac{FT}{2} \left(I + \frac{FT}{3} \left(\dots \frac{FT}{N-1} \left(I + \frac{FT}{N} \right) \dots \right) \right), \quad (4.62)$$

which has better numerical properties than the direct series of powers. We then find Γ from Eq. (4.61) and Φ from Eq. (4.60). A discussion of the selection of N and a technique to compute Ψ for comparatively large T is given by Källström (1973), and a review of various methods is found in a classic paper by Moler and Van Loan (1978). The program logic for computation of Φ and Γ for simple cases is given in Fig. 4.17. MATLAB's `c2d.m` and all control design packages that we know of compute Φ and Γ from the continuous matrices F , G , and the sample period T .

To compare this method of representing the plant with the discrete transfer functions, we can take the z -transform of Eq. (4.59) with $w = 0$ and obtain

$$\begin{aligned} [zI - \Phi]X &= \Gamma U(z), \\ Y(z) &= HX(z) \end{aligned} \quad (4.63)$$

Figure 4.17

Program logic to compute Φ and Γ from F , G , and T for simple cases. (The left arrow $\leftarrow$ is read as "is replaced by.")

-
1. Select sampling period T and description matrices F and G .
 2. Matrix $I \leftarrow$ Identity
 3. Matrix $\Psi \leftarrow I$
 4. $k \leftarrow 11$ [We are using $N = 11$ in Eq. (4.62).]
 5. If $k = 1$, go to step 9.
 6. Matrix $\Psi \leftarrow I + \frac{FT}{k} \Psi$
 7. $k \leftarrow k + 1$
 8. Go to step 5.
 9. Matrix $\Gamma \leftarrow T\Psi G$
 10. Matrix $\Phi \leftarrow I + FT\Psi$
-

therefore

$$\frac{Y(z)}{U(z)} = \mathbf{H}[z\mathbf{I} - \Phi]^{-1}\Gamma. \quad (4.64)$$

◆ **Example 4.11** Φ and Γ Calculation

By hand, calculate the Φ and Γ matrices for the satellite attitude-control system of Example 4.9.

Solution. Use Eqs. (4.60) and (4.61) and the values for $\mathbf{F}$ and $\mathbf{G}$ defined in Eq. (4.47). Since $\mathbf{F}^2 = 0$ in this case, we have

$$\begin{aligned} \Phi &= \mathbf{I} + \mathbf{F}T + \frac{\mathbf{F}^2 T^2}{2!} + \cdots \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} T = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \\ \Gamma &= \left[\mathbf{I}T + \mathbf{F}\frac{T^2}{2!} + \frac{\mathbf{F}^2 T^3}{3!} \right] \mathbf{G} \\ &= \left\{ \begin{bmatrix} T & 0 \\ 0 & T \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \frac{T^2}{2} \right\} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} T^2/2 \\ T \end{bmatrix}, \end{aligned}$$

hence, using Eq. (4.64), we obtain

$$\begin{aligned} \frac{Y(z)}{U(z)} &= [1 \quad 0] \left\{ z \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \right\}^{-1} \begin{bmatrix} T^2/2 \\ T \end{bmatrix} \\ &= \frac{T^2}{2} \frac{(z+1)}{(z-1)^2}, \end{aligned}$$

which is the same result that would be obtained using Eq. (4.41) and the z -transform tables. Note that the values for Φ and Γ could have been obtained for a specific value of T by the MATLAB statements

```
sysC = ss(F,G,H,I)
sysD = c2d(sysC,T,zoh)
[phi,gam,h,J] = ssdata(sysD)
```

Note that to compute Y/U we find that the denominator is the determinant $\det(z\mathbf{I} - \Phi)$, which comes from the matrix inverse in Eq. (4.64). This determinant is the characteristic polynomial of the transfer function, and the zeros of the determinant are the poles of the plant. We have two poles at $z = 1$ in this case, corresponding to the two integrations in this plant's equations of motion.

We can explore further the question of poles and zeros and the state-space description by considering again the transform equations (4.63). An interpretation of transfer-function poles from the perspective of the corresponding difference equation is that a pole is a value of z such that the equation has a nontrivial solution when the forcing input is zero. From Eq. (4.63a), this implies that the linear eigenvalue equations

$$[z\mathbf{I} - \Phi]\mathbf{X}(z) = [\mathbf{0}]$$

have a nontrivial solution. From matrix algebra the well-known requirement for this is that $\det(z\mathbf{I} - \Phi) = 0$. Using the Φ from the previous example, we have

$$\begin{aligned}\det[z\mathbf{I} - \Phi] &= \det \left[\begin{bmatrix} z & 0 \\ 0 & z \end{bmatrix} - \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \right] \\ &= \det \begin{bmatrix} z-1 & -T \\ 0 & z-1 \end{bmatrix} \\ &= (z-1)^2 = 0,\end{aligned}$$

which is the characteristic equation, as we have seen. To compute the poles numerically when the matrices are given, one would use an eigenvalue routine. In MATLAB, the statement

$$\text{lam}=\text{eig}(\text{phi})$$

will produce a vector, lam, of the poles of Φ .

Along the same line of reasoning, a system zero is a value of z such that the system output is zero even with a nonzero state-and-input combination. Thus if we are able to find a nontrivial solution for $X(z_0)$ and $U(z_0)$ such that $Y(z_0)$ is zero, then z_0 is a zero of the system. Combining the two parts of Eq. (4.59), we must satisfy the requirement

$$\begin{bmatrix} z\mathbf{I} - \Phi & -\Gamma \\ \mathbf{H} & 0 \end{bmatrix} \begin{bmatrix} X(z) \\ U(z) \end{bmatrix} = [\mathbf{0}]. \quad (4.65)$$

Once more the condition for the existence of nontrivial solutions is that the determinant of the square coefficient system matrix be zero.¹³ For the satellite example, we have

$$\begin{aligned}\det \begin{bmatrix} z-1 & -T & -T^2/2 \\ 0 & z-1 & -T \\ 1 & 0 & 0 \end{bmatrix} &= 1 \cdot \det \begin{bmatrix} -T & -T^2/2 \\ z-1 & -T \end{bmatrix} \\ &= +T^2 + \left(\frac{T^2}{2}\right)(z-1)\end{aligned}$$

¹³ We do not consider here the case of different numbers of inputs and outputs.

$$\begin{aligned}
&= + \frac{T^2}{2} z + \frac{T^2}{2} \\
&= + \frac{T^2}{2} (z + 1).
\end{aligned}$$

Thus we have a single zero at $z = -1$, as we have seen from the transfer function. These zeros are called **transmission zeros** and are easily computed using MATLAB's `tzero.m`.¹⁴ Using the discrete model `sysD` found in Example 4.11 the statement

```
zer=tzero(sysD)
```

produces the transmission zeros in the quantity `zer`.

4.3.4 *State-Space Models for Systems with Delay

Thus far we have discussed the calculation of discrete state models from continuous, ordinary differential equations of motion. Now we present the formulas for including a time delay in the model and also a time prediction up to one period which corresponds to the **modified z -transform** as defined by Jury. We begin with a state-variable model that includes a delay in control action. The state equations are

$$\begin{aligned}
\dot{\mathbf{x}}(t) &= \mathbf{F}\mathbf{x}(t) + \mathbf{G}u(t - \lambda), \\
y &= \mathbf{H}\mathbf{x}.
\end{aligned} \tag{4.66}$$

The general solution to Eq. (4.66) is given by Eq. (4.55); it is

$$\mathbf{x}(t) = e^{\mathbf{F}(t-t_0)}\mathbf{x}(t_0) + \int_{t_0}^t e^{\mathbf{F}(t-\tau)}\mathbf{G}u(\tau - \lambda) d\tau.$$

If we let $t_0 = kT$ and $t = kT + T$, then

$$\mathbf{x}(kT + T) = e^{\mathbf{F}T}\mathbf{x}(kT) + \int_{kT}^{kT+T} e^{\mathbf{F}(kT+T-\tau)}\mathbf{G}u(\tau - \lambda) d\tau.$$

If we substitute $\eta = kT + T - \tau$ for τ in the integral, we find a modification of Eq. (4.57)

$$\begin{aligned}
\mathbf{x}(kT + T) &= e^{\mathbf{F}T}\mathbf{x}(kT) + \int_T^0 e^{\mathbf{F}\eta}\mathbf{G}u(kT + T - \lambda - \eta)(-d\eta) \\
&= e^{\mathbf{F}T}\mathbf{x}(kT) + \int_0^T e^{\mathbf{F}\eta}\mathbf{G}u(kT + T - \lambda - \eta) d\eta.
\end{aligned}$$

¹⁴ In using this function, one must be careful to account properly for the zeros that are at infinity; the function might return them as very large numbers that the user must remove to “uncover” the finite zeros; that is, to scale the finite numbers so they don't appear to be zero by the computer.

If we now separate the system delay λ into an integral number of sampling periods plus a fraction, we can define an integer ℓ and a positive number m less than one such that

$$\lambda = \ell T - mT, \quad (4.67)$$

and

$$\begin{aligned} \ell &\geq 0, \\ 0 &\leq m < 1. \end{aligned}$$

With this substitution, we find that the discrete system is described by

$$\mathbf{x}(kT + T) = e^{\mathbf{F}T} \mathbf{x}(kT) + \int_0^T e^{\mathbf{F}\eta} \mathbf{G} u(kT + T - \ell T + mT - \eta) d\eta. \quad (4.68)$$

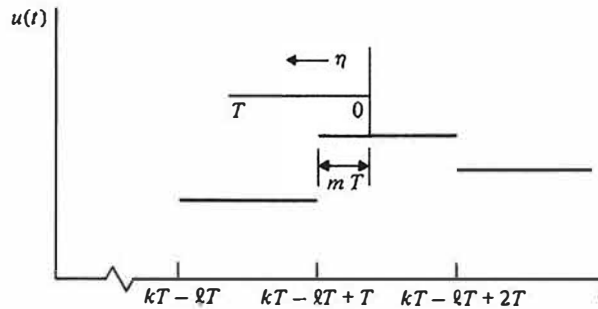
If we sketch a segment of the time axis near $t = kT - \ell T$ (Fig. 4.18), the nature of the integral in Eq. (4.68) with respect to the variable η will become clear. The integral runs for η from 0 to T , which corresponds to t from $kT - \ell T + T + mT$ backward to $kT - \ell T + mT$. Over this period, the control, which we assume is piecewise constant, takes on first the value $u(kT - \ell T + T)$ and then the value $u(kT - \ell T)$. Therefore, we can break the integral in (2.66) into two parts as follows

$$\begin{aligned} \mathbf{x}(kT + T) &= e^{\mathbf{F}T} \mathbf{x}(kT) + \int_0^{mT} e^{\mathbf{F}\eta} \mathbf{G} d\eta u(kT - \ell T + T) \\ &\quad + \int_{mT}^T e^{\mathbf{F}\eta} \mathbf{G} d\eta u(kT - \ell T) \\ &= \Phi \mathbf{x}(kT) + \Gamma_1 u(kT - \ell T) + \Gamma_2 u(kT - \ell T + T). \end{aligned} \quad (4.69)$$

In Eq. (4.69) we defined

$$\Phi = e^{\mathbf{F}T}, \quad \Gamma_1 = \int_{mT}^T e^{\mathbf{F}\eta} \mathbf{G} d\eta, \quad \text{and} \quad \Gamma_2 = \int_0^{mT} e^{\mathbf{F}\eta} \mathbf{G} d\eta. \quad (4.70)$$

Figure 4.18
Sketch of a piecewise
input and time axis for a
system with time delay



To complete our analysis it is necessary to express Eq. (4.69) in standard state-space form. To do this we must consider separately the cases of $\ell = 0$, $\ell = 1$, and $\ell > 1$.

For $\ell = 0$, $\lambda = -mT$ according to Eq. (4.67), which implies not delay but prediction. Because mT is restricted to be less than T , however, the output will not show a sample before $k = 0$, and the discrete system will be causal. The result is that the discrete system computed with $\ell = 0$, $m \neq 0$ will show the response at $t = 0$, which the same system with $\ell = 0$, $m = 0$ would show at $t = mT$. In other words, by taking $\ell = 0$ and $m \neq 0$ we pick up the response values *between* the normal sampling instants. In z -transform theory, the transform of the system with $\ell = 0$, $m \neq 0$ is called the **modified z -transform**.¹⁵ The state-variable form requires that we evaluate the integrals in Eq. (4.70). To do so we first convert Γ_1 to a form similar to the integral for Γ_2 . From Eq. (4.70) we factor out the constant matrix $\mathbf{G}$ to obtain

$$\Gamma_1 = \int_{mT}^T e^{\mathbf{F}\eta} d\eta \mathbf{G}.$$

If we set $\sigma = \eta - mT$ in this integral, we have

$$\begin{aligned} \Gamma_1 &= \int_0^{T-mT} e^{\mathbf{F}(mT+\sigma)} d\sigma \mathbf{G} \\ &= e^{\mathbf{F}mT} \int_0^{T-mT} e^{\mathbf{F}\sigma} d\sigma \mathbf{G}. \end{aligned} \quad (4.71)$$

For notational purposes we will define, for any positive nonzero scalar number, a , the two matrices

$$\Phi(a) = e^{\mathbf{F}a}, \quad \Psi(a) = \frac{1}{a} \int_0^a e^{\mathbf{F}\sigma} d\sigma. \quad (4.72)$$

In terms of these matrices, we have

$$\begin{aligned} \Gamma_1 &= (T - mT)\Phi(mT)\Psi, \\ \Gamma_2 &= mT\Psi. \end{aligned} \quad (4.73)$$

The definitions in Eqs. (4.72) are also useful from a computational point of view. If we recall the series definition of the matrix exponential

$$\Phi(a) = e^{\mathbf{F}a} = \sum_{k=0}^{\infty} \frac{\mathbf{F}^k a^k}{k!},$$

then we get

$$\begin{aligned} \Psi(a) &= \frac{1}{a} \int_0^a \sum_{k=0}^{\infty} \frac{\mathbf{F}^k \sigma^k}{k!} d\sigma \\ &= \frac{1}{a} \sum_{k=0}^{\infty} \frac{\mathbf{F}^k}{k!} \frac{a^{k+1}}{k+1} \end{aligned}$$

¹⁵ See Jury (1964) or Ogata (1987).

$$= \sum_{k=0}^{\infty} \frac{\mathbf{F}^k a^k}{(k+1)!}. \quad (4.74)$$

But now we note that the series for $\Phi(a)$ can be written as

$$\Phi(a) = \mathbf{I} + \sum_{k=1}^{\infty} \frac{\mathbf{F}^k a^k}{k!}.$$

If we let $k = j + 1$ in the sum, then, as in Eq. (4.60), we have

$$\begin{aligned} \Phi(a) &= \mathbf{I} + \sum_{j=0}^{\infty} \frac{\mathbf{F}^{j+1} a^{j+1}}{(j+1)!} \\ &= \mathbf{I} + \sum_{j=0}^{\infty} \frac{\mathbf{F}^j a^j}{(j+1)!} a \mathbf{F} \\ &= \mathbf{I} + a \Psi(a) \mathbf{F}. \end{aligned} \quad (4.75)$$

The point of Eq. (4.75) is that only the series for $\Psi(a)$ needs to be computed and from this single sum we can compute Φ and Γ .

If we return to the case $\ell = 0, m \neq 0$, the discrete state equations are

$$\mathbf{x}(k+1) = \Phi \mathbf{x}(k) + \Gamma_1 u(k) + \Gamma_2 u(k+1),$$

where Γ_1 and Γ_2 are given by Eq. (4.73). In order to put these equations in state-variable form, we must eliminate the term in $u(k+1)$. To do this, we define a new state, $\xi(k) = \mathbf{x}(k) - \Gamma_2 u(k)$. Then the equations are

$$\begin{aligned} \xi(k+1) &= \mathbf{x}(k+1) - \Gamma_2 u(k+1) \\ &= \Phi \mathbf{x}(k) + \Gamma_1 u(k) + \Gamma_2 u(k+1) - \Gamma_2 u(k+1), \\ \xi(k+1) &= \Phi[\xi(k) + \Gamma_2 u(k)] + \Gamma_1 u(k) \\ &= \Phi \xi(k) + (\Phi \Gamma_2 + \Gamma_1) u(k) \\ &= \Phi \xi(k) + \Gamma u(k). \end{aligned} \quad (4.76)$$

The output equation is

$$\begin{aligned} y(k) &= \mathbf{H} \mathbf{x}(k) \\ &= \mathbf{H}[\xi(k) + \Gamma_2 u(k)] \\ &= \mathbf{H} \xi(k) + \mathbf{H} \Gamma_2 u(k) \\ &= \mathbf{H}_d \xi(k) + J_d u(k). \end{aligned} \quad (4.77)$$

Thus for $\ell = 0$, the state equations are given by Eqs. (4.73), (4.76), and (4.77). Note especially that if $m = 0$, then $\Gamma_2 = 0$, and these equations reduce to the previous model with no delay.

Our next case is $\ell = 1$. From Eq. (4.69), the equations are given by

$$\mathbf{x}(k+1) = \Phi \mathbf{x}(k) + \Gamma_1 u(k-1) + \Gamma_2 u(k).$$

In this case, we must eliminate $u(k-1)$ from the right-hand side, which we do by defining a new state $x_{n+1}(k) = u(k-1)$. We have thus an increased dimension of the state, and the equations are

$$\begin{bmatrix} \mathbf{x}(k+1) \\ x_{n+1}(k+1) \end{bmatrix} = \begin{bmatrix} \Phi & \Gamma_1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ x_{n+1}(k) \end{bmatrix} + \begin{bmatrix} \Gamma_2 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = \begin{bmatrix} \mathbf{H} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ x_{n+1} \end{bmatrix}. \quad (4.78)$$

For our final case, we consider $\ell > 1$. In this case, the equations are

$$\mathbf{x}(k+1) = \Phi \mathbf{x}(k) + \Gamma_1 u(k-\ell) + \Gamma_2 u(k-\ell+1)$$

and we must eliminate the past controls up to $u(k)$. To do this we introduce ℓ new variables such that

$$x_{n+1}(k) = u(k-\ell), \quad x_{n+2}(k) = u(k-\ell+1), \quad \dots, \quad x_{n+\ell}(k) = u(k-1).$$

The structure of the equations is

$$\begin{bmatrix} \mathbf{x}(k+1) \\ x_{n+1}(k+1) \\ x_{n+2}(k+1) \\ \vdots \\ x_{n+\ell}(k+1) \end{bmatrix} = \begin{bmatrix} \Phi & \Gamma_1 & \Gamma_2 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ x_{n+1}(k) \\ x_{n+2}(k) \\ x_{n+3}(k) \\ \vdots \\ x_{n+\ell}(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} u(k)$$

$$y(k) = \begin{bmatrix} \mathbf{H} & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ x_{n+1}(k) \\ \vdots \\ x_{n+\ell}(k) \end{bmatrix}. \quad (4.79)$$

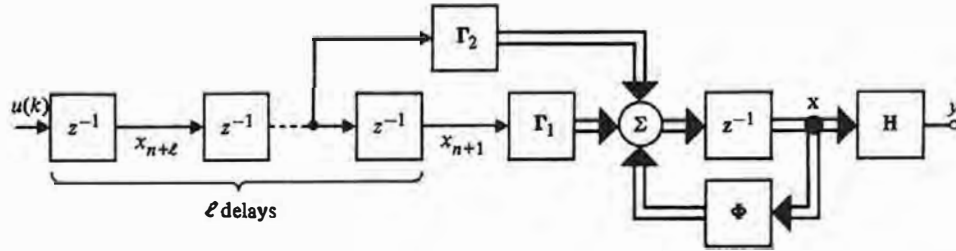
This final solution is easily visualized in terms of a block diagram, as shown in Fig. 4.19.

4.3.5 *Numerical Considerations in Computing Φ and Γ

The numerical considerations of these computations are centered in the approximation to the infinite sum for Ψ given by Eq. (4.74) or, for $a = T$, by Eq. (4.62). The problem is that if $\mathbf{F}T$ is large, then $(\mathbf{F}T)^N/N!$ becomes extremely large before it becomes small, and before acceptable accuracy is realized most computer number representations will overflow, destroying the value of the computation. Källström (1973) has analyzed a technique used by Kalman and Englar (1966),

Figure 4.19

Block diagram of system with delay of more than one period. Double line indicates vector valued variables



which has been found effective by Moler and Van Loan (1978). The basic idea comes from Eq. (4.52) with $t_2 - t_0 = 2T$ and $t_1 - t_0 = T$, namely

$$(e^{\mathbf{F}T})^2 = e^{\mathbf{F}T} e^{\mathbf{F}T} = e^{\mathbf{F}2T}. \quad (4.80)$$

Thus, if T is too large, we can compute the series for $T/2$ and square the result. If $T/2$ is too large, we compute the series for $T/4$, and so on, until we find a k such that $T/2^k$ is *not* too large. We need a test for deciding on the value of k . We propose to approximate the series for Ψ , which can be written

$$\Psi\left(\frac{T}{2^k}\right) = \sum_{j=0}^{N-1} \frac{[\mathbf{F}(T/2^k)]^j}{(j+1)!} + \sum_{j=N}^{\infty} \frac{(\mathbf{F}T/2^k)^j}{(j+1)!} = \hat{\Psi} + \mathbf{R}.$$

We will select k , the factor that decides how much the sample period is divided down, to yield a small remainder term $\mathbf{R}$. Källström suggests that we estimate the size of $\mathbf{R}$ by the size of the first term ignored in $\hat{\Psi}$, namely,

$$\hat{\mathbf{R}} \cong (\mathbf{F}T)^N / (N+1)! 2^{Nk}.$$

A simpler method is to select k such that the size of $\mathbf{F}T$ divided by 2^k is less than 1. In this case, the series for $\mathbf{F}T/2^k$ will surely converge. The rule is to select k such that

$$2^k > \|\mathbf{F}T\| = \max_j \sum_{i=1}^n |F_{ij}| T.$$

Taking the log of both sides, we find

$$k > \log_2 \|\mathbf{F}T\|,$$

from which we select

$$k = \max(\lceil \log_2 \|\mathbf{F}T\|, 0), \quad (4.81)$$

where the symbol $\lceil x$ means the smallest integer greater than x . The maximum of this integer and zero is taken because it is possible that $\|FT\|$ is already so small that its log is negative, in which case we want to select $k = 0$.

Having selected k , we now have the problem of computing $\hat{\Psi}(T)$ from $\hat{\Psi}(T/2^k)$. Our original concept was based on the series for Φ , which satisfied Eq. (4.80). To obtain the suitable formula for Ψ , we use the relation between Φ and Ψ given by Eq. (4.60) as follows to obtain the “doubling” formula for Ψ

$$\begin{aligned}\Phi(2T) &= \Phi(T)\Phi(T), \\ \mathbf{I} + 2TF\Psi(2T) &= [\mathbf{I} + TF\Psi(T)][\mathbf{I} + TF\Psi(T)] \\ &= \mathbf{I} + 2TF\Psi(T) + T^2F^2\Psi^2(T);\end{aligned}$$

therefore

$$2TF\Psi(2T) = 2TF\Psi(T) + T^2F^2\Psi^2(T).$$

This is equivalent to

$$\Psi(2T) = \left(\mathbf{I} + \frac{TF}{2}\Psi(T) \right) \Psi(T),$$

which is the form to be used. The program logic for computing Ψ is shown in Fig. 4.20.¹⁶ This algorithm does not include the delay discussed in Section 4.3.4.

Figure 4.20

Logic for a program to compute Ψ using automatic time scaling

-
1. Select F and T .
 2. Comment: Compute $\|FT\|$.
 3. $V \leftarrow \max_j \{ \sum_i |F_{ij}| \} \times T$
 4. $k \leftarrow$ smallest nonnegative integer greater than $\log_2 V$.
 5. Comment: compute $\Psi(T/2^k)$.
 6. $T_1 \leftarrow T/2^k$
 7. $\mathbf{I} \leftarrow$ Identity
 8. $\Psi \leftarrow \mathbf{I}$
 9. $j \leftarrow 1$
 10. If $j = 1$, go to step 14.
 11. $\Psi \leftarrow \mathbf{I} + \frac{FT_1}{j}\Psi$
 12. $j \leftarrow j - 1$
 13. Go to step 10.
 14. Comment: Now double Ψ k times.
 15. If $k = 0$, stop.
 16. $\Psi \leftarrow \left(\mathbf{I} + \frac{FT}{2^{k+1}}\Psi \right) \Psi$
 17. $k \leftarrow k - 1$
 18. Go to step 15.
-

¹⁶ Similar logic is used by MATLAB in C2D.M.

For that, we must implement the logic shown in Fig. 4.19. In the Control Toolbox, the function `c2d.m` executes the logic with a delay if one is specified.

4.3.6 *Nonlinear Models

Contrary to the predominant developments in this book, models of dynamic systems are generally nonlinear. However, it is more difficult to apply analysis to nonlinear models and, thus, less insight is gained if models are left in their nonlinear form throughout the entire design process. Controls engineers commonly use numerical simulation of nonlinear models to evaluate the performance of control systems, a technique that should always be a part of any control system design. To aid in the design synthesis of controllers and to gain insight into approximate behavior, it is often advantageous to linearize the system so the methods in this text can be utilized.

We begin with the assumption that our plant dynamics are adequately described by a set of ordinary differential equations in state-variable form as

$$\begin{aligned} \dot{x}_1 &= f_1(x_1, \dots, x_n, u_1, \dots, u_m, t), \\ \dot{x}_2 &= f_2(x_1, \dots, x_n, u_1, \dots, u_m, t), \\ &\vdots \\ \dot{x}_n &= f_n(x_1, \dots, x_n, u_1, \dots, u_m, t), \\ y_1 &= h_1(x_1, \dots, x_n, u_1, \dots, u_m, t), \\ &\vdots \\ y_p &= h_p(x_1, \dots, x_n, u_1, \dots, u_m, t), \end{aligned} \quad (4.82)$$

or, more compactly in matrix notation, we assume that our plant dynamics are described by

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{f}(\mathbf{x}, \mathbf{u}, t), \\ \mathbf{x}(t_0) &= \mathbf{x}_0, \\ \mathbf{y} &= \mathbf{h}(\mathbf{x}, \mathbf{u}, t). \end{aligned} \quad (4.83)$$

One proceeds as follows with the process of linearization and small-signal approximations. We assume stationarity by the approximation that $\mathbf{f}$ and $\mathbf{h}$ do not change significantly from their initial values at t_0 . Thus we can set

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t_0)$$

or, simply

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}), \quad \mathbf{y} = \mathbf{h}(\mathbf{x}, \mathbf{u}). \quad (4.84)$$

The assumption of small signals can be reflected by taking $\mathbf{x}$ and $\mathbf{u}$ to be always close to their reference values $\mathbf{x}_0$, $\mathbf{u}_0$, and these values, furthermore, to be an equilibrium point of Eq. (4.82), where

$$\mathbf{f}(\mathbf{x}_0, \mathbf{u}_0) = 0. \quad (4.85)$$

Now, if $\mathbf{x}$ and $\mathbf{u}$ are “close” to $\mathbf{x}_0$ and $\mathbf{u}_0$, they can be written as $\mathbf{x} = \mathbf{x}_0 + \delta\mathbf{x}$; $\mathbf{u} = \mathbf{u}_0 + \delta\mathbf{u}$, and these can be substituted into Eq. (4.84). The fact that $\delta\mathbf{x}$ and $\delta\mathbf{u}$ are small is now used to motivate an expansion of Eq. (4.84) about $\mathbf{x}_0$ and $\mathbf{u}_0$ and to suggest that the only terms in the first power of the small quantities $\delta\mathbf{x}$ and $\delta\mathbf{u}$ need to be retained. We thus have a vector equation and need the expansion of a vector-valued function of a vector variable,

$$\frac{d}{dt}(\mathbf{x}_0 + \delta\mathbf{x}) = \mathbf{f}(\mathbf{x}_0 + \delta\mathbf{x}, \mathbf{u}_0 + \delta\mathbf{u}). \quad (4.86)$$

If we go back to Eq. (4.82) and do the expansion of the components f_i one at a time, it is tedious but simple to verify that Eq. (4.86) can be written as¹⁷

$$\delta\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}_0, \mathbf{u}_0) + \mathbf{f}_{,x}(\mathbf{x}_0, \mathbf{u}_0)\delta\mathbf{x} + \mathbf{f}_{,u}(\mathbf{x}_0, \mathbf{u}_0)\delta\mathbf{u} + \cdots \quad (4.87)$$

where we define the partial derivative of a scalar f_i with respect to the vector $\mathbf{x}$ by a subscript notation:

$$f_{i,x} \triangleq \left(\frac{\partial f_i}{\partial x_1} \quad \cdots \quad \frac{\partial f_i}{\partial x_n} \right). \quad (4.88)$$

The row vector in Eq. (4.88) is called the **gradient** of the scalar f_i with respect to the vector $\mathbf{x}$. If $\mathbf{f}$ is a vector, we define its partial derivatives with respect to the vector $\mathbf{x}$ as the matrix (called the **Jacobian**) composed of *rows* of gradients. In the subscript notation, if we mean to take the partial of *all* components, we omit the specific subscript such as 1 or 2 but hold its place by the use of a comma

$$\mathbf{f}_{,x} = \begin{bmatrix} \frac{\partial f_1}{\partial \mathbf{x}} \\ \frac{\partial f_2}{\partial \mathbf{x}} \\ \vdots \\ \frac{\partial f_n}{\partial \mathbf{x}} \end{bmatrix}. \quad (4.89)$$

Now, to return to Eq. (4.87), we note that by Eq. (4.85) we chose $\mathbf{x}_0$, $\mathbf{u}_0$ to be an equilibrium point, so the first term on the right of Eq. (4.87) is zero, and because the terms beyond those shown depend on higher powers of the small signals $\delta\mathbf{x}$ and $\delta\mathbf{u}$, we are led to the approximation

$$\begin{aligned} \delta\dot{\mathbf{x}} &\approx \mathbf{f}_{,x}(\mathbf{x}_0, \mathbf{u}_0)\delta\mathbf{x} + \mathbf{f}_{,u}(\mathbf{x}_0, \mathbf{u}_0)\delta\mathbf{u}, \\ \delta\mathbf{y} &= \mathbf{h}_{,x}\delta\mathbf{x} + \mathbf{h}_{,u}\delta\mathbf{u}. \end{aligned} \quad (4.90)$$

¹⁷ Note that $d\mathbf{x}_0/dt = 0$ because our “reference trajectory” $\mathbf{x}_0$ is a constant here.

But now the notation is overly clumsy, so we drop the $\delta\mathbf{x}$, $\delta\mathbf{u}$ and $\delta\mathbf{y}$ notation and simply call them $\mathbf{x}$, $\mathbf{u}$ and $\mathbf{y}$ and define the constant matrices

$$\begin{aligned}\mathbf{F} &= \mathbf{f}_{,x}(\mathbf{x}_0, \mathbf{u}_0), & \mathbf{G} &= \mathbf{f}_{,u}(\mathbf{x}_0, \mathbf{u}_0), \\ \mathbf{H} &= \mathbf{h}_{,x}(\mathbf{x}_0, \mathbf{u}_0), & \mathbf{J} &= \mathbf{h}_{,u}(\mathbf{x}_0, \mathbf{u}_0).\end{aligned}$$

This results in the form we used earlier in Section 2.1.1

$$\dot{\mathbf{x}} = \mathbf{F}\mathbf{x} + \mathbf{G}\mathbf{u}, \quad \mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{J}\mathbf{u}. \quad (4.91)$$

We go even further and restrict ourselves to the case of single input and single output and discrete time. We then write the model as

$$\begin{aligned}\mathbf{x}(k+1) &= \mathbf{\Phi}\mathbf{x}(k) + \mathbf{\Gamma}u(k), \\ y(k) &= \mathbf{H}\mathbf{x}(k) + Ju(k),\end{aligned} \quad (4.92)$$

from which the transfer function is

$$G(z) = Y(z)/U(z) = \mathbf{H}(z\mathbf{I} - \mathbf{\Phi})^{-1}\mathbf{\Gamma} + J. \quad (4.93)$$

Thus we see that nonlinear models can be approximated as linear state-space models or as transfer function models. The accuracy of the approximation varies with the problem, but is generally useful in designing the control system. The final design of the control system should always be checked via numerical simulation of the nonlinear equations.

4.4 Signal Analysis and Dynamic Response

In Section 4.2 we demonstrated that if two variables are related by a linear constant difference equation, then the ratio of the z -transform of the output signal to that of the input is a function of the system equation alone, and the ratio is called the transfer function. A method for study of linear constant discrete systems is thereby indicated, consisting of the following steps:

1. Compute the transfer function of the system $H(z)$.
2. Compute the transform of the input signal, $E(z)$.
3. Form the product, $E(z)H(z)$, which is the transform of the output signal, U .
4. Invert the transform to obtain $u(kT)$.

If the system description is available in difference-equation form, and if the input signal is elementary, then the first three steps of this process require very little effort or computation. The final step, however, is tedious if done by hand, and, because we will later be preoccupied with design of transfer functions to give desirable responses, we attach great benefit to gaining intuition for the kind of response to be expected from a transform without actually inverting it or numerically evaluating the response. Our approach to this problem is to present a

repertoire of elementary signals with known features and to learn their representation in the transform or z -domain. Thus, when given an unknown transform, we will be able, by reference to these known solutions, to infer the major features of the time-domain signal and thus to determine whether the unknown is of sufficient interest to warrant the effort of detailed time-response computation. To begin this process of attaching a connection between the time domain and the z -transform domain, we compute the transforms of a few elementary signals.

4.4.1 The Unit Pulse

We have already seen that the unit pulse is defined by¹⁸

$$\begin{aligned} e_1(k) &= 1 & (k = 0) \\ &= 0 & (k \neq 0) \\ &= \delta_k; \end{aligned}$$

therefore we have

$$E_1(z) = \sum_{-\infty}^{\infty} \delta_k z^{-k} = z^0 = 1. \quad (4.94)$$

This result is much like the continuous case, wherein the Laplace transform of the unit impulse is the constant 1.0.

The quantity $E_1(z)$ gives us an instantaneous method to relate signals to systems: To characterize the system $H(z)$, consider the signal $u(k)$, which is the unit pulse response; then $U(z) = H(z)$.

4.4.2 The Unit Step

Consider the unit step function defined by

$$\begin{aligned} e_2(k) &= 1 & (k \geq 0) \\ &= 0 & (k < 0) \\ &\triangleq 1(k). \end{aligned}$$

In this case, the z -transform is

$$\begin{aligned} E_2(z) &= \sum_{k=-\infty}^{\infty} e_2(k) z^{-k} = \sum_{k=0}^{\infty} z^{-k} \\ &= \frac{1}{1 - z^{-1}} & (|z^{-1}| < 1) \\ &= \frac{z}{z - 1} & (|z| > 1). \end{aligned} \quad (4.95)$$

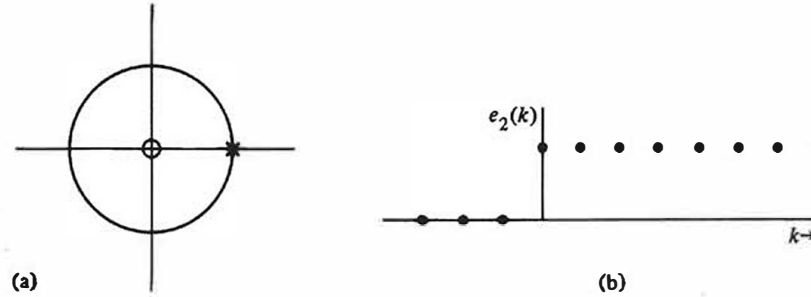
¹⁸ We have shifted notation here to use $e(k)$ rather than e_k for the k th sample. We use subscripts to identify different signals.

Here the transform is characterized by a zero at $z = 0$ and a pole at $z = 1$. The significance of the convergence being restricted to $|z| > 1$ will be explored later when we consider the inverse transform operation. The Laplace transform of the unit step is $1/s$; we may thus keep in mind that a pole at $s = 0$ for a continuous signal corresponds in some way to a pole at $z = 1$ for discrete signals. We will explore this further later. In any event, we record that a pole at $z = 1$ with convergence outside the unit circle, $|z| = 1$, will correspond to a constant for positive time and zero for negative time.

To emphasize the connection between the time domain and the z -plane, we sketch in Fig. 4.21 the z -plane with the unit circle shown and the pole of $E_2(z)$ marked $\times$ and the zero marked $\circ$. Beside the z -plane, we sketch the time plot of $e_2(k)$.

Figure 4.21

(a) Pole and zero of $E_2(z)$ in the z -plane. The unit circle is shown for reference. (b) Plot of $e_2(k)$



4.4.3 Exponential

The one-sided exponential in time is

$$\begin{aligned} e_3(k) &= r^k & (k \geq 0) \\ &= 0 & (k < 0) \end{aligned} \quad (4.96)$$

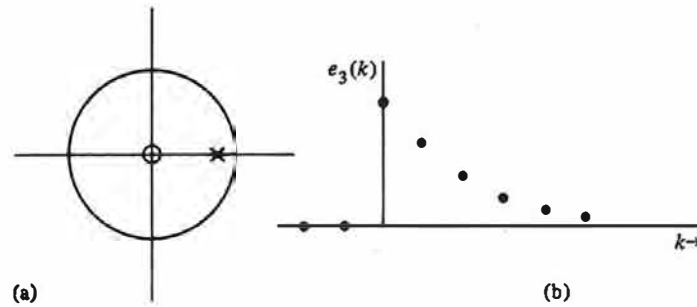
which is the same as $r^k l(k)$, using the symbol $l(k)$ for the unit step function. Now we get

$$\begin{aligned} E_3(z) &= \sum_{k=0}^{\infty} r^k z^{-k} \\ &= \sum_{k=0}^{\infty} (rz^{-1})^k \\ &= \frac{1}{1 - rz^{-1}} & (|rz^{-1}| < 1) \\ &= \frac{z}{z - r} & (|z| > |r|). \end{aligned} \quad (4.97)$$

The pole of $E_3(z)$ is at $z = r$. From Eq. (4.96) we know that $e_3(k)$ grows without bound if $|r| > 1$. From Eq. (4.97) we conclude that a z -transform that converges for large z and has a real pole *outside* the circle $|z| = 1$ corresponds to a growing signal. If such a signal were the unit-pulse response of our system, such as our digital control program, we would say the program was *unstable* as we saw in Eq. (4.39). We plot in Fig. 4.22 the z -plane and the corresponding time history of $E_3(z)$ as $e_3(k)$ for the stable value, $r = 0.6$.

Figure 4.22

(a) Pole and zero of $E_3(z)$ in the z -plane. (b) Plot of $e_3(k)$



4.4.4 General Sinusoid

Our next example considers the modulated sinusoid $e_4(k) = [r^k \cos(k\theta)]1(k)$, where we assume $r > 0$. Actually, we can decompose $e_4(k)$ into the sum of two complex exponentials as

$$e_4(k) = r^k \left(\frac{e^{jk\theta} + e^{-jk\theta}}{2} \right) 1(k),$$

and because the z -transform is linear,¹⁹ we need only compute the transform of each single complex exponential and add the results later. We thus take first

$$e_5(k) = r^k e^{jk\theta} 1(k) \quad (4.98)$$

and compute

$$\begin{aligned} E_5(z) &= \sum_{k=0}^{\infty} r^k e^{jk\theta} z^{-k} \\ &= \sum_{k=0}^{\infty} (r e^{j\theta} z^{-1})^k \end{aligned}$$

¹⁹ We have not shown this formally. The demonstration, using the definition of linearity given above, is simple and is given in Section 4.6.

$$\begin{aligned}
 &= \frac{1}{1 - r e^{j\theta} z^{-1}} \\
 &= \frac{z}{z - r e^{j\theta}} \quad (|z| > r). \quad (4.99)
 \end{aligned}$$

The signal $e_5(k)$ grows without bound as k gets large if and only if $r > 1$, and a system with this pulse response is BIBO stable if and only if $|r| < 1$. The boundary of stability is the unit circle. To complete the argument given before for $e_4(k) = r^k \cos k\theta 1(k)$, we see immediately that the other half is found by replacing θ by $-\theta$ in Eq. (4.99)

$$\mathcal{Z}\{r^k e^{-j\theta k} 1(k)\} = \frac{z}{z - r e^{-j\theta}} \quad (|z| > r), \quad (4.100)$$

and thus that

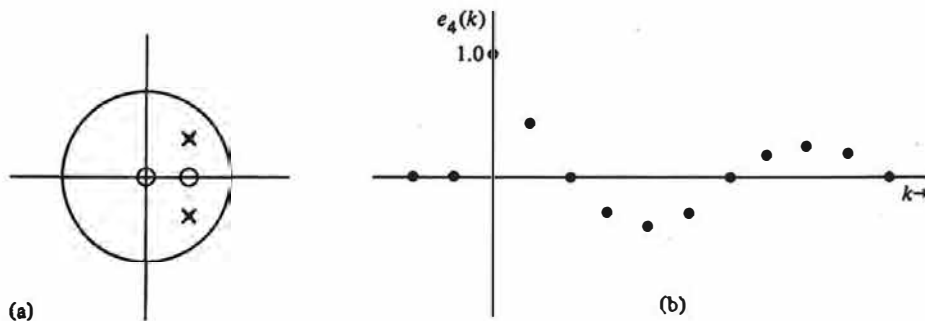
$$\begin{aligned}
 E_4(z) &= \frac{1}{2} \left\{ \frac{z}{z - r e^{j\theta}} + \frac{z}{z - r e^{-j\theta}} \right\} \\
 &= \frac{z(z - r \cos \theta)}{z^2 - 2r(\cos \theta)z + r^2} \quad (|z| > r). \quad (4.101)
 \end{aligned}$$

The z -plane pole-zero pattern of $E_4(z)$ and the time plot of $e_4(k)$ are shown in Fig. 4.23 for $r = 0.7$ and $\theta = 45^\circ$.

We note in passing that if $\theta = 0$, then e_4 reduces to e_3 and, with $r = 1$, to e_2 , so that three of our signals are special cases of e_4 . By exploiting the features of $E_4(z)$, we can draw a number of conclusions about the relation between pole locations in the z -plane and the time-domain signals to which the poles correspond. We collect these for later reference.

Figure 4.23

(a) Poles and zeros of $E_4(z)$ for $\theta = 45^\circ$, $r = 0.7$ in the z -plane. (b) Plot of $e_4(k)$



1. The settling time of a transient, defined as the time required for the signal to decay to one percent of its maximum value, is set mainly by the value of the radius, r , of the poles.

- (a) $r > 1$ corresponds to a growing signal that will not decay at all.
- (b) $r = 1$ corresponds to a signal with constant amplitude (which is *not* BIBO stable as a pulse response).
- (c) For $r < 1$, the closer r is to 0 the shorter the settling time. The corresponding system is BIBO stable. We can compute the settling time in samples, N , in terms of the pole radius, r .

Pole Radius r	Response Duration N
0.9	43
0.8	21
0.6	9
0.4	5

- (d) A pole at $r = 0$ corresponds to a transient of finite duration.

2. The number of samples per oscillation of a sinusoidal signal is determined by θ . If we require $\cos(\theta k) = \cos(\theta(k + N))$, we find that a period of 2π rad contains N samples, where

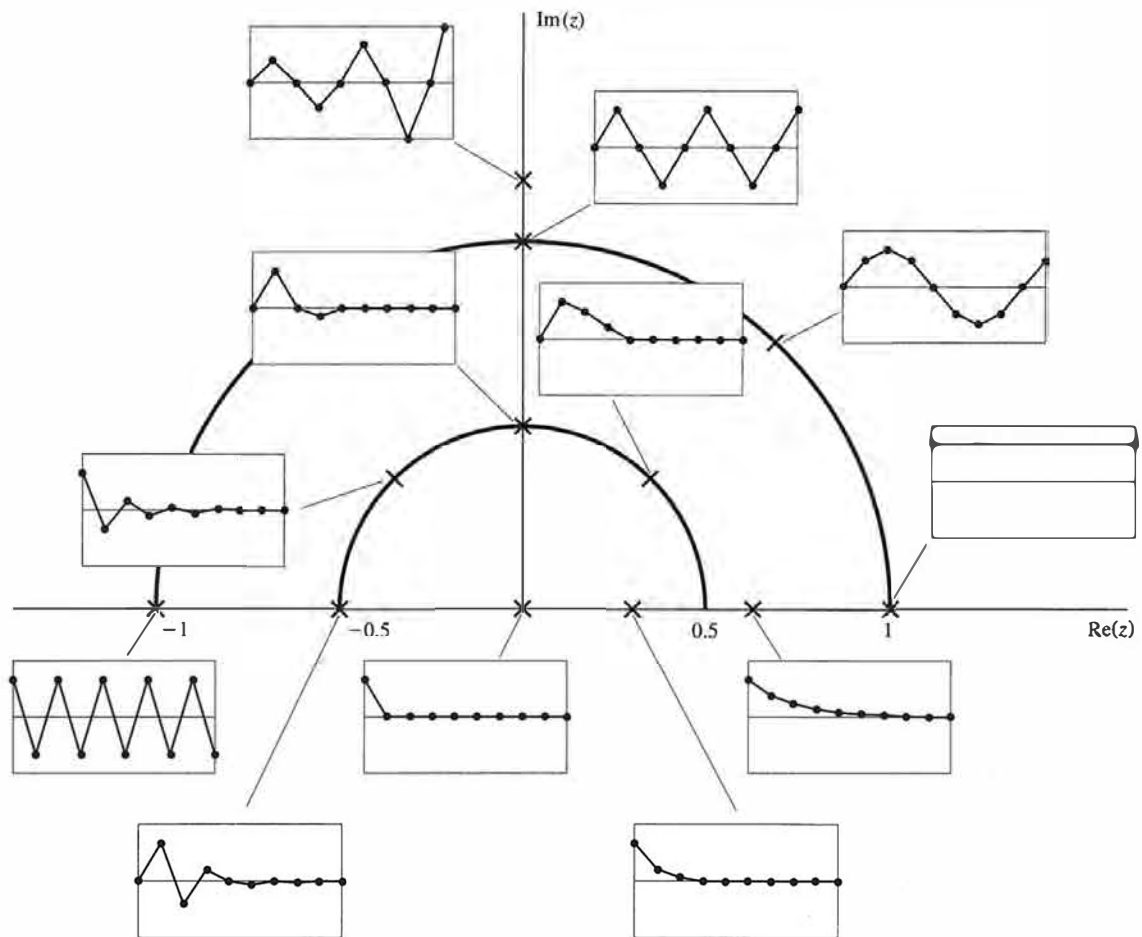
$$N = \frac{2\pi}{\theta} \Big|_{\text{rad}} = \frac{360}{\theta} \Big|_{\text{deg}} \text{ samples/cycle.}$$

For $\theta = 45^\circ$, we have $N = 8$, and the plot of $e_4(k)$ given in Fig. 4.23(b) shows the eight samples in the first cycle very clearly. A sketch of the unit circle with several points corresponding to various numbers of samples per cycle marked is drawn in Fig. 4.24 along with other contours that will be explained in the next section. The sampling frequency in Hertz is $1/T$, and the signal frequency is $f = 1/NT$ so that $N = f_s/f$ and $1/N$ is a *normalized* signal frequency. Since $\theta = (2\pi)/N$, θ is the normalized signal frequency in radians/sample. θ/T is the frequency in radians/second.

A compilation of signal responses versus their pole location in the z -plane is shown in Fig. 4.25. It demonstrates visually the features just summarized for the general sinusoid, which encompasses all possible signals.

Figure 4.25

Time sequences associated with pole locations in the z-plane



And, taking samples one second apart ($T = 1$), we have

$$\begin{aligned}
 y(kT) &= (e^{-0.3567})^k \cos \frac{\pi k}{4} 1(k) \\
 &= (0.7)^k \cos \frac{\pi k}{4} 1(k) \\
 &= e_4(k).
 \end{aligned}$$

The poles of the Laplace transform of $y(t)$ (in the s -plane) are at

$$s_{1,2} = -a + jb, -a - jb.$$

From Eq. (4.101), the z -transform of $E_4(z)$ has poles at

$$z_{1,2} = re^{j\theta}, re^{-j\theta},$$

but because $y(kT)$ equals $e_4(k)$, it follows that

$$\begin{aligned} r &= e^{-aT}, & \theta &= bT \\ z_{1,2} &= e^{s_1T}, e^{s_2T}. \end{aligned}$$

If $E(z)$ is a ratio of polynomials in z , which will be the case if $e(k)$ is generated by a linear difference equation with constant coefficients, then by partial fraction expansion, $E(z)$ can be expressed as a sum of elementary terms like E_4 and E_3 .²⁰ In all such cases, the discrete signal can be generated by samples from continuous signals where the relation between the s -plane poles and the corresponding z -plane poles is given by

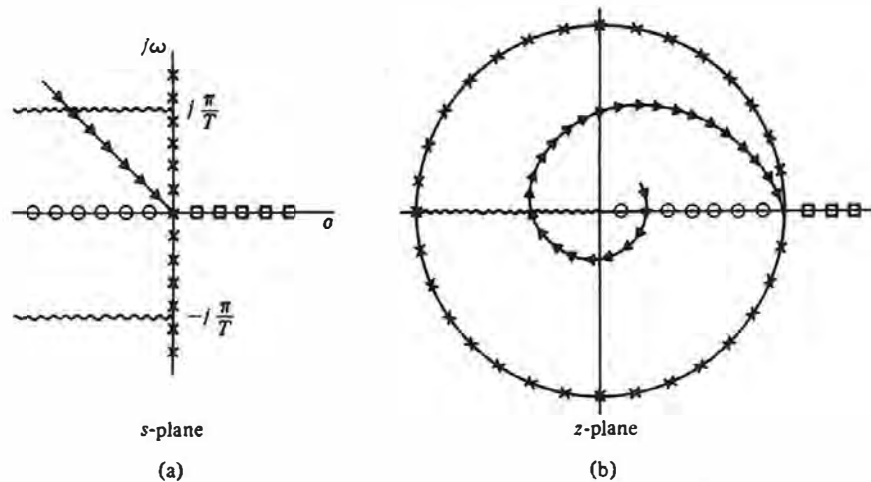
$$z = e^{sT}. \quad (4.103)$$

If we know what it means to have a pole in a certain place in the s -plane, then Eq. (4.103) shows us where to look in the z -plane to find a representation of discrete samples having the *same time features*. It is useful to sketch several major features from the s -plane to the z -plane according to Eq. (4.103) to help fix these ideas. Such a sketch is shown in Fig. 4.26.

Each feature should be traced in the mind to obtain a good grasp of the relation. These features are given in Table 4.2. We note in passing that the map

Figure 4.26

Corresponding lines in the s -plane and the z -plane according to $z = e^{sT}$



²⁰ Unless a pole of $E(z)$ is repeated. We have yet to compute the discrete version of a signal corresponding to a higher-order pole. The result is readily shown to be a polynomial in k multiplying $r^k e^{jk\theta}$.

Table 4.2 Description of corresponding lines in s -plane and z -plane

s -plane	Symbol	z -plane
$s = j\omega$ Real frequency axis $s = \sigma \geq 0$ $s = \sigma \leq 0$	$\times \times \times$ $\square \square \square$ $\circ \circ \circ$	$ z = 1$ Unit circle $z = r \geq 1$ $z = r, 0 \leq r \leq 1$
$s = -\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2}$ $= -a + jb$ Constant damping ratio if ζ is fixed and ω_n varies $s = \pm j(\pi/T) + \sigma$	$\triangle \triangle \triangle$ $\sigma \leq 0$	$z = re^{j\theta}$ where $r = \exp(-\zeta\omega_n T)$ $= e^{-aT}$ $\theta = \omega_n T\sqrt{1-\zeta^2} = bT$ Logarithmic spiral $z = -r$

$z = e^{sT}$ of Eq. (4.103) is many-to-one. There are many values of s for each value of z . In fact, if

$$s_2 = s_1 + j \frac{2\pi}{T} N,$$

then $e^{s_1 T} = e^{s_2 T}$. The (great) significance of this fact will be explored in Chapter 5.

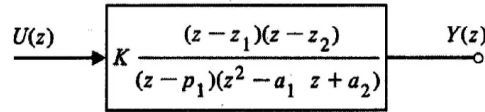
Lines of constant damping in the s -plane are mapped into the z -plane according to Eq. (4.103) for several values of ζ in Fig. 4.24. We often refer to the damping of a pole in the z -plane in terms of this **equivalent s -plane** damping, or sometimes we simply refer to the damping of a z -plane pole. Likewise, lines of constant natural frequency, ω_n , in the s -plane (semi-circles centered at the origin) are also mapped into the z -plane according to Eq. (4.103) for several values of ω_n in Fig. 4.24. It's interesting to note that in the immediate vicinity of $z = +1$, the map of ζ and ω_n looks exactly like the s -plane in the vicinity of $s = 0$. Because of the usefulness of this mapping, the Control System Toolbox has the function `zgrid.m` that allows one to superimpose this mapping on various plots to help in the interpretation of the results. You will see its use in the figure files of discrete root loci in Chapter 7.

4.4.6 Step Response

Our eventual purpose, of course, is to design digital controls, and our interest in the relation between z -plane poles and zeros and time-domain response comes from our need to know how a proposed design will respond in a given dynamic situation. The generic dynamic test for controls is the step response, and we will conclude this discussion of discrete system dynamic response with an examination of the relationships between the pole-zero patterns of elementary systems and the corresponding step responses for a discrete transfer function from u to y of a hypothetical plant. Our attention will be restricted to the step responses of the discrete system shown in Fig. 4.27 for a selected set of values of the parameters.

Figure 4.27

Definition of the parameters of the system whose step responses are to be catalogued



Note that if $z_1 = p_1$, the members of the one pole-zero pair cancel out; and if at the same time $z_2 = r \cos(\theta)$, $a_1 = -2r \cos(\theta)$, and $a_2 = r^2$, the system response, $Y(z)$, to the input with transform $U(z) = 1$ (a unit pulse) is

$$Y(z) = \frac{z - r \cos \theta}{z^2 - 2r \cos \theta z + r^2}. \quad (4.104)$$

This transform, when compared with the transform $E_4(z)$ given in Eq. (4.101), is seen to be

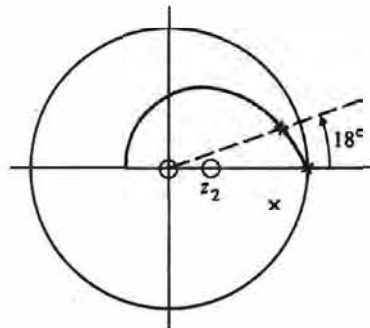
$$Y(z) = z^{-1} E_4(z),$$

and we conclude that under these circumstances the system pulse response is a delayed version of $e_4(k)$, a typical second-order system pulse response.

For our first study we consider the effect of zero location. We let $z_1 = p_1$ and explore the effect of the (remaining) zero location, z_2 , on the step-response overshoot for three sets of values of a_1 and a_2 . We select a_1 and a_2 so that the poles of the system correspond to a response with an equivalent s -plane damping ratio $\zeta = 0.5$ and consider values of θ of 18, 45, and 72 degrees. In every case, we will take the gain K to be such that the steady-state output value equals the step size. The situation in the z -plane is sketched in Fig. 4.28 for $\theta = 18^\circ$. The curve for $\zeta = 0.5$ is also shown for reference. In addition to the two poles and one zero of $H(z)$, we show the pole at $z = 1$ and the zero at $z = 0$, which come from the transform of the input step, $U(z)$, given by $z/(z - 1)$.

Figure 4.28

Pole-zero pattern of $Y(z)$ for the system of Fig. 4.27, with $z_1 = p_1$, $U(z) = z/(z - 1)$, a_1 and a_2 selected for $\theta = 18^\circ$, and $\zeta = 0.5$



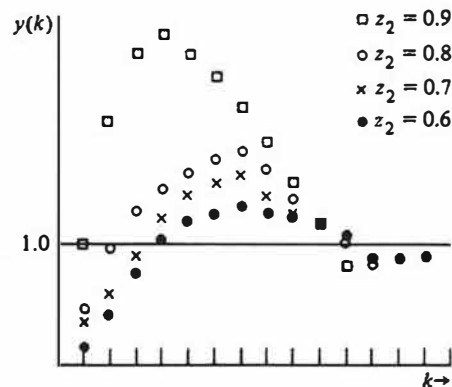
The major effect of the zero z_2 on the step response $y(k)$ is to change the percent overshoot, as can be seen from the four step responses for this case plotted in Fig. 4.29. To summarize all these data, we plot the percent overshoot versus zero location in Fig. 4.30 for $\zeta = 0.5$ and in Fig. 4.31. for $\zeta = 0.707$. The major feature of these plots is that the zero has very little influence when on the negative axis, but its influence is dramatic as it comes near $+1$. Also included on the plots of Fig. 4.30 are overshoot figures for a zero in the unstable region on the positive real axis. These responses go in the *negative* direction at first, and for the zero very near $+1$, the negative peak is larger than 1!²¹

Our second class of step responses corresponds to a study of the influence of a third pole on a basically second-order response. For this case we again consider the system of Fig. 4.27, but this time we fix $z_1 = z_2 = -1$ and let p_1 vary from near -1 to near $+1$. In this case, the major influence of the moving singularity is on the rise time of the step response. We plot this effect for $\theta = 18, 45$, and 72 degrees and $\zeta = 0.5$ on Fig. 4.32. In the figure we defined the rise time as the time required for the response to rise to 0.95 , which is to 5% of its final value. We see here that the extra pole causes the rise time to get very much longer as the location of p_1 moves toward $z = +1$ and comes to dominate the response.

Our conclusions from these plots are that the addition of a pole or zero to a given system has only a small effect if the added singularities are in the range from 0 to -1 . However, a zero moving toward $z = +1$ greatly increases the system overshoot. A pole placed toward $z = +1$ causes the response to slow down and thus primarily affects the rise time, which is progressively increased.

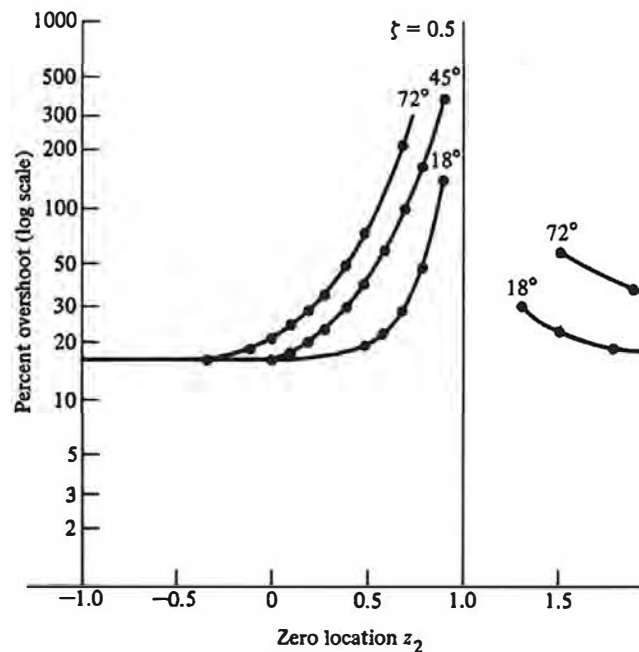
The understanding of how poles and zeros affect the time response is very useful for the control system designer. The knowledge helps guide the iterative

Figure 4.29
Plot of step responses for a discrete plant described by the pole-zero pattern of Fig. 4.28 for various values of z_2



²¹ Such systems are called nonminimum phase by Bode because the phase shift they impart to a sinusoidal input is greater than the phase of a system whose *magnitude* response is the same but that has a zero in the stable rather than the unstable region.

Figure 4.30
Effects of an extra zero
on a discrete
second-order system,
 $\zeta = 0.5$; $\theta = 18^\circ$, 45° ,
and 72°



design process and helps the designer understand why a response is the way it is. Ultimately, however, the test of a design is typically the actual time response, either by numerical simulation or an experimental evaluation. Today, transform inversion would never be carried out. In MATLAB, the numerical simulation of the impulse response for a discrete system, `sysD` is accomplished by

```
y = impulse(sysD)
```

and the discrete step response by

```
y = step(sysD)
```

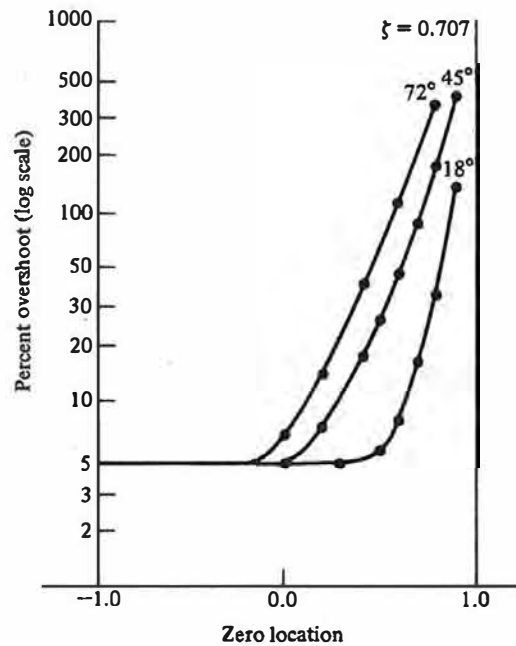
Invoked without a left-hand argument (`y =`), both functions result in a plot of the response on the screen.

4.5 Frequency Response

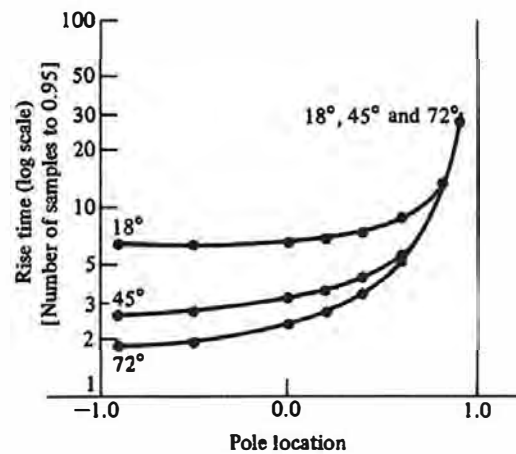
A very important concept in linear systems analysis is the frequency response. If a sinusoid at frequency ω_o is applied to a stable, linear, constant, continuous system, the response is a transient plus a sinusoidal steady state at the *same frequency*, ω_o , as the input. If the transfer function is written in gain-phase form as $H(j\omega) = A(\omega)e^{j\psi(\omega)}$, then the steady-state response to a unit-amplitude

Figure 4.31

Effects of an extra zero on a discrete second-order system, $\zeta = 0.707$; $\theta = 18^\circ, 45^\circ$, and 72° ; percent overshoot versus zero location

**Figure 4.32**

Effects of an extra pole on rise time for a discrete third-order system, two zeros at -1 , one zero at ∞ ; $\zeta = 0.5$; $\theta = 18^\circ, 45^\circ, 72^\circ$



sinusoidal signal has amplitude $A(\omega_o)$ and phase $\psi(\omega_o)$ relative to the input signal.

We can say almost exactly the same respecting the frequency response of a stable, linear, constant, discrete system. If the system has a transfer function $H(z)$,

we define its magnitude and phase for z taking on values around the unit circle by $H(e^{j\omega T}) = A(\omega T)e^{j\psi(\omega T)}$. If a unit-amplitude sinusoid is applied, then in the steady state, the response samples will be on a sinusoid of the same frequency with amplitude $A(\omega_o T)$ and phase $\psi(\omega_o T)$. It is worthwhile going through the calculations to fix ideas on this point.

From Eq. (4.16), the discrete response transform is

$$U(z) = H(z)E(z). \quad (4.105)$$

If $e(k) = \cos(\omega_o T k)1(k)$, then, from Eq. (4.101) with $r = 1$ and $\theta = \omega_o T$, we have

$$E(z) = \frac{1}{2} \left\{ \frac{z}{z - e^{-j\omega_o T}} + \frac{z}{z - e^{j\omega_o T}} \right\}. \quad (4.106)$$

If we substitute Eq. (4.106) into Eq. (4.105), we obtain

$$U(z) = \frac{1}{2} \left\{ \frac{zH(z)}{z - e^{-j\omega_o T}} + \frac{zH(z)}{z - e^{j\omega_o T}} \right\}. \quad (4.107)$$

The steady state of $u(kT)$ corresponds to the terms in the expansion of Eq. (4.107) associated with the two poles on the unit circle. If we expand $U(z)/z$ into partial fractions and multiply back by z , the steady state part can be found as

$$U_{ss}(z) = \frac{1}{2} \frac{H(e^{j\omega_o T})z}{z - e^{j\omega_o T}} + \frac{1}{2} \frac{H(e^{-j\omega_o T})z}{z - e^{-j\omega_o T}}.$$

If $H(e^{j\omega_o T}) = A(\omega_o T)e^{j\psi(\omega_o T)}$, then we have

$$U_{ss}(z) = \frac{A}{2} \frac{e^{j\psi} z}{z - e^{-j\omega_o T}} + \frac{A}{2} \frac{e^{-j\psi} z}{z - e^{j\omega_o T}}, \quad (4.108)$$

and the inverse transform of $U_{ss}(z)$ is

$$\begin{aligned} U_{ss}(kT) &= \frac{A}{2} e^{j\psi} e^{j\omega_o T k} + \frac{A}{2} e^{-j\psi} e^{-j\omega_o T k} \\ &= A \cos(\omega_o T k + \psi), \end{aligned} \quad (4.109)$$

which, of course, are samples at kT instants on a sinusoid of amplitude A , phase ψ , and frequency ω_o .

We will defer the plotting of particular frequency responses until later chapters (see, for example, Figs. 6.3, 6.8, 7.16, and 7.28). However, it should be noticed here that although a sinusoid of frequency ω_o *could* be passed through the samples of Eq. (4.109), there are other continuous sinusoids of frequency $\omega_o + \ell 2\pi/T$ for integer ℓ which also pass through these points. This is the phenomenon of aliasing, to which we will return in Chapter 5. Here, we define the discrete frequency response of a transfer function $H(z)$ to sinusoids of frequency ω_o as $H(e^{j\omega_o T})$ so that the amplitude A and phase ψ are

$$A = |H(e^{j\omega_o T})| \quad \text{and} \quad \psi = \angle(H(e^{j\omega_o T})) \quad (4.110)$$

which can be evaluated and plotted by MATLAB's `bode.m` with the scripts

```
sysD = tf(num,den,T)
bode(sysD)
```

where amplitude is plotted in decibels (dB), or

```
[mag,phase,w] = bode(sysD)
subplot(2,1,1), loglog(w,mag)
subplot(2,1,2), semilogx(w,phase)
```

where amplitude is plotted as a ratio as in the figures in this text. If the system is described by the state-space matrices, the scripts above can be invoked with

```
sysD = ss(F,G,H,J,T).
```

4.5.1 *The Discrete Fourier Transform (DFT)

The analysis developed above based on the z -transform is adequate for considering the theoretical frequency response of a linear, constant system or the corresponding difference equation, but it is not the best for the analysis of real-time signals as they occur in the laboratory or in other experimental situations. For the analysis of real data, we need a transform defined over a finite data record, which can be computed quickly and accurately. The required formula is that of the **Discrete Fourier Transform**, the DFT, and its numerical cousin, the **Fast Fourier Transform**, the FFT. Implementation of a version of the FFT algorithm is contained in all signal-processing software and in most computer-aided control-design software.

Fast Fourier Transform

To understand the DFT, it is useful to consider two properties of a signal and its Fourier transform that are complements of each other: the property of being periodic and the property of being discrete. In ordinary Fourier analysis, we have a signal that is neither periodic nor discrete and its Fourier transform is also neither discrete nor periodic. If, however, the time function $f(t)$ is periodic with period T_0 , then the appropriate form of the transform is the Fourier series, and the transform is defined only for the discrete frequencies $\omega = 2\pi n/T_0$. In other words, if the function in time is periodic, the function in frequency is discrete. The case where the properties are reversed is the z -transform we have just been studying. In this case, the time functions are discrete, being sampled, and the z -transform is periodic in ω ; for if $z = e^{j\omega T}$, corresponding to real frequencies, then replacing $\omega = \omega + 2\pi k/T$ leaves z unchanged. We can summarize these results with the following table:

	Time	Frequency
Fourier Series	periodic	discrete
z-transform	discrete	periodic

Suppose we now have a time function that is both periodic and discrete. Based on what we have seen, we would expect the transform of this function also to be both periodic and discrete. And this is the case, which leads us to the finite discrete Fourier transform and its finite inverse. Let the time function in question be $f(kT) = f(kT + NT)$. Because the function is periodic, the transform can be defined as the finite sum

$$F\left(\frac{2\pi n}{NT}\right) = \sum_{k=0}^{N-1} f(kT) e^{-j2\pi(nkT)/(NT)}.$$

This is the same as the z-transform over one period evaluated at the discrete frequencies of a Fourier series $\omega = 2\pi n/NT$. It is standard practice to suppress all the arguments except the indices of time and frequency and write

$$F_n = \sum_{k=0}^{N-1} f_k e^{-j2\pi(nk)/N}. \quad (4.111)$$

To complete the DFT, we need the inverse transform, which, by analogy with the standard Fourier transform, we guess to be the sum

$$\sum_{n=0}^{N-1} F_n e^{j2\pi(nk)/N}.$$

If we substitute Eq. (4.111) with summing index ℓ into this, we find

$$\sum_{n=0}^{N-1} \left\{ \sum_{\ell=0}^{N-1} f_{\ell} e^{-j2\pi(n\ell)/N} \right\} e^{j2\pi(nk)/N}.$$

Interchanging the order of the summations gives

$$\sum_{\ell=0}^{N-1} f_{\ell} \left\{ \sum_{n=0}^{N-1} e^{j2\pi n(k-\ell)/N} \right\}.$$

The sum in the braces is a finite geometric series, which we can evaluate as follows

$$\begin{aligned} \sum_{n=0}^{N-1} e^{j2\pi n(k-\ell)/N} &= \frac{1 - e^{j2\pi(k-\ell)}}{1 - e^{j2\pi(k-\ell)/N}} \\ &= \begin{cases} N & k - \ell = 0 \\ 0 & k - \ell = 1, 2, \dots, N-1. \end{cases} \end{aligned}$$

The sum is periodic with period N . With this evaluation, we see that the sum we have been considering is Nf_k , and thus we have the inverse sum

$$f_k = \frac{1}{N} \sum_{n=0}^{N-1} F_n e^{j2\pi(nk)/N}. \quad (4.112)$$

Equations (4.111) and (4.112) comprise the DFT

$$F_n = \sum_{k=0}^{N-1} f_k e^{-j2\pi(nk)/N},$$

$$f_k = \frac{1}{N} \sum_{n=0}^{N-1} F_n e^{j2\pi(nk)/N}.$$

Because there are N terms in the sum in Eq. (4.111), it would appear that to compute the DFT for one frequency it will take on the order of N multiply and add operations; and to compute the DFT for all N frequencies, it would take on the order of N^2 multiply and add operations. However, several authors, especially Cooley and Tukey (1965), have showed how to take advantage of the circular nature of the exponential so that all N values of F_n can be computed with the order of $N \log(N)$ operations if N is a power of 2. For $N = 1024$, this is a saving of a factor of 100, a very large value. Their algorithm and related schemes are called the Fast Fourier Transform or FFT.

To use the DFT/FFT in evaluating frequency response, consider a system described by Eq. (4.105) and where the input is a sinusoid at frequency $\omega_\ell = 2\pi\ell/NT$ so that $e(kT) = A \sin(2\pi\ell kT/NT)$. We apply this input to the system *and wait until all transients have died away*. At this time, the output is given by $u(kT) = B \sin(2\pi\ell k/N + \psi)$. The DFT of $e(k)$ is

$$E_n = \sum_{k=0}^{N-1} A \sin\left(\frac{2\pi\ell k}{N}\right) e^{-j(2\pi nk)/N}$$

$$= \sum_{k=0}^{N-1} \frac{A}{2j} [e^{j(2\pi\ell k)/N} - e^{-j(2\pi\ell k)/N}] e^{-j(2\pi nk)/N}$$

$$= \begin{cases} 0, & \ell \neq n \\ \frac{NA}{2j}, & \ell = n. \end{cases}$$

The DFT of the output is

$$U_n = \sum_{k=0}^{N-1} B \sin\left(\frac{2\pi\ell k}{N} + \psi\right) e^{-j(2\pi nk)/N}$$

$$= \sum_{k=0}^{N-1} \frac{B}{2j} [e^{j\psi} e^{j(2\pi\ell k)/N} - e^{-j\psi} e^{-j(2\pi\ell k)/N}] e^{-j(2\pi nk)/N}$$